

Algebra: Multiplying Complex Numbers

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Name: _____ Date: _____ Score: _____

DIRECTIONS

Multiply the complex numbers. Use $i^2 = -1$ to simplify. Write your answer in $a + bi$ form.

1 Multiply:

$$3i \cdot 5i$$

Answer: _____

2 Multiply:

$$2i \cdot (3+4i)$$

Answer: _____

3 Multiply:

$$(1+i)(1-i)$$

Answer: _____

4 Multiply:

$$(2+3i)(1+i)$$

Answer: _____

5 Multiply:

$$(3-2i)(3+2i)$$

Answer: _____

6 Multiply:

$$(1+2i)(2+3i)$$

Answer: _____

7 Multiply:

$$(2+i)^2$$

Answer: _____

8 Multiply:

$$(4-i)(2+3i)$$

Answer: _____

9 Multiply:

$$(3+4i)(3-4i)$$

Answer: _____

10 Multiply:

$$(1-i)(2-3i)$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

Distribute like polynomials (FOIL for two binomials). Replace i^2 with -1 . Combine real parts and imaginary parts separately.

tely.

1 $3i \cdot 5i$
 $= -15$

$$15i^2 = 15(-1) = -15$$

2 $2i(3+4i)$
 $= -8 + 6i$

$$6i+8i^2 = 6i-8 = -8+6i$$

3 $(1+i)(1-i)$
 $= 2$

$$1-i^2 = 1+1 = 2$$

4 $(2+3i)(1+i)$
 $= -1 + 5i$

$$2+2i+3i+3i^2 = 2+5i-3 = -1+5i$$

5 $(3-2i)(3+2i)$
 $= 13$

$$9-4i^2 = 9+4 = 13$$

6 $(1+2i)(2+3i)$
 $= -4 + 7i$

$$2+3i+4i+6i^2 = 2+7i-6 = -4+7i$$

7 $(2+i)^2$
 $= 3 + 4i$

$$4+4i+i^2 = 4+4i-1 = 3+4i$$

8 $(4-i)(2+3i)$
 $= 11 + 10i$

$$8+12i-2i-3i^2 = 8+10i+3 = 11+10i$$

9 $(3+4i)(3-4i)$
 $= 25$

$$9-16i^2 = 9+16 = 25$$

10 $(1-i)(2-3i)$
 $= -1 - 5i$

$$2-3i-2i+3i^2 = 2-5i-3 = -1-5i$$

